High-Response Operation by Ultra-Low Inertia Servo Motors

The product lines include HK-MT series (small capacity, ultra-low inertia) and HK-RT series (medium capacity, ultra-low inertia). The ultra-low inertia servo motors enable a high-response operation that reduces the cycle time of an ultra-high-throughput material handling system.

Compact, High-Power Rate Servo Motors for High-Speed Operation

Medium-capacity HK-RT series 1 to 7 kW

Comparison of HG-RR (previous series) and HK-RT in 1 kW

( ): Increased torque

Servo motor model		HG-RR103	HK-RT103W	
Rated output of a combined servo amplifier	[kW]	2.0	**1.0 (2.0)**	Smaller capacity servo amplifier
Flange size	[mm]	100	**90**	Reduced flange size (by 10 %)
Rated torque	[N·m]	3.2		
Maximum torque	[N·m]	8.0	**8.0 (9.5)**	Increased torque (to 118 %)
Maximum speed	[r/min]	4500	**6700**	Increased speed (to 148 %)
Moment of inertia J	[× 10^{-4} kg·m^2]	1.50	**0.721**	Lower inertia (by 52 %)
Power rate at rated torque	[kW/s]	67.4	**141**	Increased responsivity (to 209 %)
Motor length	[mm]	145.5	**118.9**	Reduced motor length (by 26.6 mm)

Comparison of HK-KT (low inertia) and HK-RT in 2 kW

( ): Increased torque

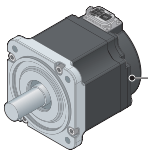
Servo motor model		HK-KT203W	HK-RT203W	
Flange size	[mm]	90		
Rated torque	[N·m]	6.4		
Maximum torque	[N·m]	19.1 (25.5)	15.9 (19.1)	
Maximum speed	[r/min]	6000	**6700**	Increased speed (to 111 %)
Moment of inertia J	[× 10^{-4} kg·m^2]	5.65	**1.28**	Lower inertia (by 77 %)
Power rate at rated torque	[kW/s]	71.7	**317**	Increased responsivity (to 442 %)
Motor length	[mm]	136.9	172.9	

Maximum Speed of 10000 r/min

Small-capacity HK-MT series 0.05 to 1 kW

The high-power rate servo motors are optimal for packaging machines and material handling systems. Servo motors with maximum speed of 10000 r/min *1 are added to the product lines, contributing to a shorter cycle time.

*1. The high-speed type models have "V" in the model name and are equipped with an incremental encoder.



Maximum speed
Standard servo motor: 6700 r/min
High-speed servo motor: 10000 r/min *1

Model Designation for 1-Axis Servo Amplifier (Note 1)

B | B-RJ

M R - J 5 - 1 0 B _ - _ _

AC servo amplifier MELSERVO-J5 series

Symbol	Interface
B	SSCNET III/H

Symbol	Power supply
None	3-phase 200 V AC, 1-phase 200 V AC (Note 5), or DC input
4	3-phase 400 V AC

Symbol	Rated output [kW]
10	0.1
20	0.2
40	0.4
60	0.6
70	0.75
100	1
200	2
350	3.5
500	5
700	7
12K	12
17K	17
25K	25

Symbol	Special specifications
None	Standard
RJ	Fully closed loop control four-wire type, Load-side encoder A/B/Z-phase input compatible
ED	MR-J5-_B_ without a dynamic brake (Note 3)
RU	MR-J5-_B_-RJ without a dynamic brake (Note 3)
PX	MR-J5-_B_ without an enclosed regenerative resistor (Note 4)
RZ	MR-J5-_B_-RJ without an enclosed regenerative resistor (Note 4)
LL	MR-J5-_B_ compatible with pressure control
RN	MR-J5-_B_-LL without an enclosed regenerative resistor (Note 4)
EB	MR-J5-_B_ with special coating specification (Note 6)
RB	MR-J5-_B_-RJ with special coating specification (Note 6)

M R - J 5 - 1 0 A _ - _ _

A | A-RJ

AC servo amplifier MELSERVO-J5 series

Symbol	Interface
A	General-purpose

Symbol	Power supply
None	3-phase 200 V AC, 1-phase 200 V AC (Note 5), or DC input
4	3-phase 400 V AC

Symbol	Rated output [kW]
10	0.1
20	0.2
40	0.4
60	0.6
70	0.75
100	1
200	2
350	3.5
500	5
700	7
12K	12
17K	17
25K	25

Symbol	Special specifications (Note 2)
None	Standard
RJ	Fully closed loop control four-wire type, Load-side encoder A/B/Z-phase input compatible, High resolution analog input compatible
ED	MR-J5-_A_ without a dynamic brake (Note 3)
RU	MR-J5-_A_-RJ without a dynamic brake (Note 3)
PX	MR-J5-_A_ without an enclosed regenerative resistor (Note 4)
RZ	MR-J5-_A_-RJ without an enclosed regenerative resistor (Note 4)
EB	MR-J5-_A_ with special coating specification (Note 6)
RB	MR-J5-_A_-RJ with special coating specification (Note 6)

Notes:
1. This section describes what each symbol in a model name indicates. Some combinations of symbols are not available.
2. For the servo amplifier firmware version supporting each function, refer to "MR-J5 User's Manual".
3. A dynamic brake which is built in the 7 kW or smaller servo amplifiers is removed. When the servo amplifiers without the dynamic brake are used, the servo motors coast to a stop and do not stop immediately at alarm occurrence or power failure. Take measures to ensure safety on the entire system. When specified servo motors are used, the electronic dynamic brake may activate at an alarm occurrence. The dynamic brake can be disabled with a servo parameter setting. Refer to "MR-J5 User's Manual" for details.
4. Available in 12 kW to 25 kW servo amplifiers. A regenerative resistor (standard accessory) is not enclosed. Refer to "MR-J5 User's Manual" for details.
5. A power supply of 1-phase 200 V AC is supported by 0.1 kW to 2 kW servo amplifiers.
6. The special coating is applied to the circuit board of the servo amplifier. Refer to "MR-J5 User's Manual" for details.

Positioning Function: Point Table Method

G | G-HS | G-RJ | WG | DG | MDG

Absolute value command method: travels to a specified address (absolute value) with reference to the home position

Item	Setting range	Description
Point table No.	1 to 255	Specify a point table in which a target position, servo motor speed, acceleration/deceleration, acceleration time constant/deceleration time constant, dwell, auxiliary function, and M code will be set.
Target position (Note 1) (position data)	-2147483.648 to 2147483.647 [mm] -214748.3648 to 214748.3647 [inch] -360.000 to 360.000 [degree] -2147483648 to 2147483647 [pulse]	Set a travel distance. (1) When using as absolute position command method Set a target address (absolute value). (2) When using as relative position command method Set a travel distance. Reverse rotation command is applied with a minus sign.
Servo motor speed (Note 2)	0 to maximum speed [r/min] 0 to 2147483.647 [mm/s] 0 to 214748.3647 [inch/s] 0 to 2147483.647 [degree/s] 0 to 2147483647 [pulse/s]	Set a command speed for the servo motor in positioning.
Acceleration	0 to 2147483.647 [mm/s^2] 0 to 214748.3647 [inch/s^2] 0 to 2147483.647 [degree/s^2] 0 to 2147483647 [pulse/s^2]	Set an acceleration for the servo motor to reach the set speed. (Acceleration time [s] = Servo motor speed/Acceleration)
Acceleration time constant	0 to 20000 [ms]	Set a time period for the servo motor to reach the rated speed.
Deceleration	0 to 2147483.647 [mm/s^2] 0 to 214748.3647 [inch/s^2] 0 to 2147483.647 [degree/s^2] 0 to 2147483647 [pulse/s^2]	Set a deceleration for the servo motor to decelerate from the set speed to a stop. (Deceleration time [s] = Servo motor speed/Deceleration)
Deceleration time constant	0 to 20000 [ms]	Set a time period for the servo motor to decelerate from the set speed to a stop.
Dwell	0 to 20000 [ms]	Set a dwell. When the dwell is set, the position command for the next point table will be started after the position command for the selected point table is completed and the set dwell is passed. The dwell is disabled when the auxiliary function is set to 0 or 2. Continuous operation is enabled when the auxiliary function is set to 1, 3, 8, 9, 10, or 11 and the dwell is set to 0.
Auxiliary function	0 to 3, 8 to 11	Set auxiliary function. (1) When using the point table with the absolute position command method 0: Automatic operation for a selected point table is performed. 1: Automatic operation is performed to the next point table. 8: Automatic operation for a point table selected at startup is performed. 9: Automatic operation of the point table No. 1 is performed. (2) When using the point table with the relative position command method 2: Automatic operation for a selected point table is performed. 3: Automatic operation is performed to the next point table. 10: Automatic operation for a point table selected at startup is performed. 11: Automatic operation of the point table No. 1 is performed.
M code	0 to 255	Set a code to be outputted when the positioning is complete.

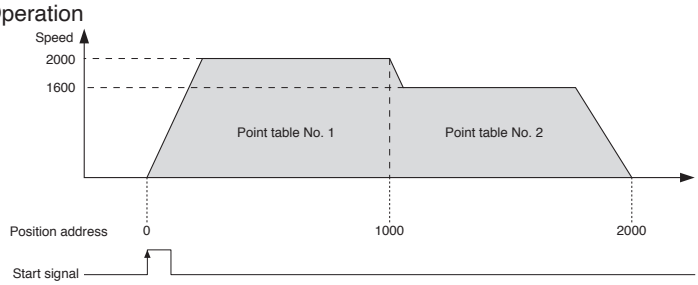
Notes: 1. Change the unit to mm/inch/degree/pulse with [Pr. PT01].
2. The speed unit is r/min for the rotary servo motors and the direct drive motors, and mm/s for the linear servo motors.

Example of setting point table data

Point table example

Point table No.	Target position (position data)	Servo motor speed [r/min]	Acceleration time constant [ms]	Deceleration time constant [ms]	Dwell [ms]	Auxiliary function	M code
1	1000	2000	200	200	0	1	1
2	2000	1600	100	100	0	0	2
:	:	:	:	:	:	:	:
255	3000	3000	100	100	0	2	99

Operation



MR-CV_ Specifications (Note 3) (400 V)

DG

Power regeneration converter unit model MR-CV_		11K4	18K4	30K4	37K4	45K4	55K4	75K4
Output	Rated voltage	513 V DC to 648 V DC						
	Rated current [A]	21	38	72	82	99	119	150
Main circuit power supply input	Voltage/frequency (Note 1)	3-phase 380 V AC to 480 V AC, 50 Hz/60 Hz						
	Rated current [A]	18	35	61	70	85	106	130
	Permissible voltage fluctuation	3-phase 323 V AC to 528 V AC						
	Permissible frequency fluctuation	±3 % maximum						
Control circuit power supply input	Voltage/frequency	1-phase 380 V AC to 480 V AC, 50 Hz/60 Hz						
	Rated current [A]	0.1						
	Permissible voltage fluctuation	1-phase 323 V AC to 528 V AC						
	Permissible frequency fluctuation	±3 % maximum						
	Power consumption [W]	30						
Interface power supply		24 V DC ± 10 % (required current capacity: 0.35 A)						
Capacity [kW]		11	18	30	37	45	55	75
Protective functions		Undervoltage protection, regenerative error protection, regenerative overvoltage shut-off, MC drive circuit error protection, open-phase detection, inrush current suppression circuit error protection, main circuit device overheat error protection, cooling fan error protection, overload shut-off (electronic thermal)						
Continuous rating [kW]		7.5	11	20	25		55	
Instantaneous maximum rating [kW]		39	60	92	101	125	175	180
Structure (IP rating)		Force cooling, open (IP20) (Note 2)						
Mass [kg]		6.1		12.1			25.0	

Notes: 1. Rated output and speed of a rotary servo motor are applicable when the power regeneration converter unit is operated within the specified power supply voltage and frequency.
2. Terminal blocks are excluded.
3. MR-CV_4 power regeneration converter units require a mounting attachment. Refer to "Mounting Attachment" in this catalog for details.

MR-CV_ Connection Example

For the connection example of power regeneration converter units, refer to "Main/Control Circuit Power Supply Connection Example For connecting MR-CV_ and MR-J5D_-G4(-N1)" in this catalog.

Selection of Converter Unit, Servo Amplifier, and Drive Unit

Combination of a simple converter and servo amplifiers

G G-HS G-RJ WG B B-RJ WB A A-RJ

Select a servo amplifier for connection that meets the following conditions.

(1) For MR-CM3K

- Connectable servo amplifier models
  MR-J5-10_ to MR-J5-200_, MR-J5W2-22_ to MR-J5W2-1010_, MR-J5W3-222_/MR-J5W3-444_
- The sum of rated capacities [kW] of connected servo amplifiers ≤ 3 kW (MR-CM3K rated output)
  When using a multi-axis servo amplifier, calculate the sum of the rated capacities of all axes as the rated capacity of the servo amplifier.
- Number of connectable servo amplifiers to one MR-CM3K ≤ 6
  When connecting a multi-axis servo amplifier, count each servo amplifier unit as one unit, rather than the number of axes.

(2) For MR-CM08K1

- Connectable servo amplifier models
  MR-J5-10_ to MR-J5-70_, MR-J5W2-22_, MR-J5W2-44_, MR-J5W3-222_
- The sum of rated capacities [kW] of connected servo amplifiers ≤ 0.8 kW (MR-CM08K1 rated output)
  When using a multi-axis servo amplifier, calculate the sum of the rated capacities of all axes as the rated capacity of the servo amplifier.
- Number of connectable servo amplifier axes to one MR-CM08K1 ≤ 3 axes
  When connecting a multi-axis servo amplifier, count the number of connectable servo amplifier axes.

	MR-CM3K (200 V)	MR-CM08K1
Maximum number of connectable servo amplifiers/axes	6	3 axes (counted as 2 axes for MR-J5W2-_ and 3 axes for MR-J5W3-_)
Total capacity of connectable servo amplifiers	3 kW	0.8 kW
Continuous rating	3 kW	0.8 kW
Instantaneous maximum rating	9 kW	1.5 kW

Combination of a power regeneration converter unit and drive units

DG

Select a power regeneration converter unit which meets the following conditions. When all the conditions are satisfied, multiple MR-J5D_-G4(-N1) drive units can be connected to one power regeneration converter unit. When connecting the multiple MR-J5D_-G4(-N1) drive units, install the drive units in descending order of capacity per axis from the right side of the power regeneration converter unit.
Refer to "MR-J5D User's Manual" for details of the selection.

(1) Effective value [kW] of total output power of servo motors ≤ Continuous rating [kW] of MR-CV_
(2) Maximum value [kW] of total output power of servo motors × 1.2 ≤ Instantaneous maximum rating [kW] of MR-CV_
(3) Total widths of MR-J5D_-G4(-N1) (one side) ≤ 1500 mm

	MR-CV_ (400 V)						
	11K4	18K4	30K4	37K4	45K4	55K4	75K4
Continuous rating [kW]	7.5	11	20	25	25	55	55
Instantaneous maximum rating [kW]	39	60	92	101	125	175	180
Total widths of MR-J5D_-G4(-N1)	1500 mm or shorter						

	MR-J5D1-_(-N1)					MR-J5D2-_(-N1)					MR-J5D3-_(-N1)	
	100G4	200G4	350G4	500G4	700G4	100G4	200G4	350G4	500G4	700G4	100G4	200G4
Unit width [mm]	60					60			75		60	

HK-KT Series Recommended Load to Motor Inertia Ratio

The values in brackets are the servo motor speed. (Example: "≤ 6000 r/min" = 6000 r/min or less, "> 6000 r/min" = Exceeding 6000 r/min)
Contact your local sales office if the load to motor inertia ratio exceeds the value in the table.

Rotary servo motor		HK-KT	When connected with a 200 V servo amplifier	When connected with a 400 V servo amplifier	When connected with a 400 V drive unit
HK-KT_W	40 × 40	053W	20 times or less (Note 1)	20 times or less	20 times or less
		13W	20 times or less (Note 1)	20 times or less	20 times or less
		1M3W	20 times or less	20 times or less	20 times or less
	60 × 60	13UW	10 times or less (Note 1)	-	-
		23W	28 times or less (≤ 6000 r/min) 23 times or less (> 6000 r/min)	-	-
		43W	23 times or less	-	-
		63W	25 times or less	-	-
	80 × 80	23UW	10 times or less	-	-
		43UW	10 times or less	-	-
		7M3W	16 times or less	-	-
		103W	17 times or less	-	-
	90 × 90	63UW	10 times or less	-	-
		7M3UW	10 times or less	-	-
		103UW	15 times or less	-	-
		153W	15 times or less	-	-
		203W	15 times or less	-	-
		202W	15 times or less	-	-
HK-KT_4_W	60 × 60	434W	25 times or less	23 times or less	23 times or less
		634W	25 times or less	30 times or less (≤ 3000 r/min) 20 times or less (> 3000 r/min)	30 times or less
	80 × 80	7M34W	17 times or less	20 times or less (≤ 3000 r/min) 9 times or less (> 3000 r/min)	20 times or less
		1034W	17 times or less	30 times or less (≤ 3000 r/min) 7 times or less (> 3000 r/min)	30 times or less
	90 × 90	634UW	-	10 times or less	10 times or less
		1034UW	-	10 times or less	10 times or less
		1534W	15 times or less	30 times or less (≤ 3000 r/min) 11 times or less (> 3000 r/min)	10 times or less
		2034W	15 times or less	30 times or less (≤ 3000 r/min) 10 times or less (> 3000 r/min)	9 times or less
		2024W	15 times or less	15 times or less	15 times or less

Rotary servo motor	When connected with MR-MD333G(-N1)
HK-KT0536E2-S1	30 times or less (Note 2)

Notes: 1. When the servo motor is combined with a 0.1 kW servo amplifier, this recommended load to motor inertia ratio is applicable for operating the servo motor at the rated speed. If operating speed exceeds the rated speed, check whether a regenerative option is required using drive sizing software Motorizer. A servo amplifier with a larger capacity can be combined.
2. MR-MD333G(-N1) is not equipped with a built-in regenerative resistor. When the servo motor speed is high, an overvoltage alarm may occur. In this case, review the operation pattern to lower the load to motor inertia ratio and the servo motor speed.

List of Linear Encoders (Note 1)

Mitsubishi Electric high-speed serial communication-compatible absolute type

Manufacturer	Model	Resolution	Rated speed (Note 2)	Maximum effective measurement length (Note 3)	Communication method (Note 4)
Magnescale Co., Ltd.	SR77	0.01 µm, 0.05 µm	3.3 m/s	2040 mm	Two-wire type
	SR87			3040 mm	
	SR27A	0.01 µm	3.3 m/s	2040 mm	Two-wire type/ Four-wire type
	SR67A			3640 mm	
	SmartSCALE SQ47	0.005 µm	3.3 m/s	3740 mm	
	SmartSCALE SQ57			3770 mm	
Mitutoyo Corporation	AT343A	0.05 µm	2.0 m/s	3000 mm	Two-wire type
	AT543A-SC		2.5 m/s	2200 mm	
	AT545A-SC	20 µm/4096 (Approx. 0.005 µm)	2.5 m/s	2200 mm	
	ST743A	0.1 µm	5.0 m/s	6000 mm	
	ST744A				
	ST748A				
	ST1341A	0.01 µm	8.0 m/s	12000 mm	
	ST1342A	0.001 µm		4200 mm	
	ABS AT1343A (Note 6)	0.05 µm	3.0 m/s	2200 mm	
	ABS AT1344A (Note 6)	0.01 µm			
	ABS AT1347A (Note 6)	0.001 µm			
Renishaw	RESOLUTE RL40M	1 nm	100.0 m/s	2100 mm	Two-wire type
		50 nm		20990 mm	
	EVOLUTE EL40M	50 nm, 100 nm, 500 nm	100.0 m/s	10010 mm	
	FORTiS-S 40N	1 nm, 10 nm, 50 nm	4.0 m/s	3240 mm (Note 5)	Four-wire type
	FORTiS-N 40N				
Heidenhain	LC 495M	0.001 µm, 0.01 µm	3.0 m/s	2040 mm	Four-wire type
	LC 496M				
	LC 195M			4240 mm (Note 5)	
	LC 196M				
	LIC 4193M	0.001 µm, 0.005 µm, 0.01 µm	10.0 m/s	3040 mm	Two-wire type/ Four-wire type
	LIC 4195M			28440 mm	
	LIC 4197M			6040 mm	
	LIC 4199M			1020 mm	
	LIC 3197M	0.01 µm	10.0 m/s	10000 mm	
	LIC 3199M				
	LIC 2197M	0.05 µm, 0.1 µm	10.0 m/s	6020 mm	
	LIC 2199M			6020 mm	
	MC15M	0.05 µm, 0.1 µm	10.0 m/s	3020 mm	
Nidec Instruments Corporation	PSLH206 (Note 6)	0.1 µm	5.0 m/s	1300 mm	Two-wire type
	PSLH323 (Note 6)			2000 mm	
	PSLH326 (Note 6)	0.2 µm	5.0 m/s	10000 mm	
Nidec Machine Tool Corporation	MPFA-HZ-M01	0.1 µm	30.0 m/s	8000 mm	Two-wire type
FAGOR AUTOMATION	L2AMH (Note 6)	0.01 µm, 0.05 µm	8.0 m/s	3020 mm	Two-wire type
	G2AM (Note 6)	0.01 µm, 0.05 µm	3.0 m/s	3040 mm	

Notes:
1. Contact the relevant linear encoder manufacturer for details on operating environment and specifications of the linear encoder such as ambient temperature, vibration resistance and IP rating.
2. The listed values are the manufacturer's specifications. When combined with servo amplifiers, the specification value is either the listed value or the servo motor maximum speed, whichever is lower.
3. The listed values are the manufacturer's specifications for linear encoder effective measurement lengths. The maximum length of the encoder cable between a linear encoder and a servo amplifier is 30 m. For a linear encoder manufactured by Nidec Machine Tool Corporation, the maximum length of the encoder cable between the linear encoder and a servo amplifier is 20 m.
4. The compatible communication method varies by the servo amplifier and operation mode. Refer to "External Encoder Connection Specifications" in this catalog.
5. When the resolution is 1 nm, the maximum effective measurement length is 2040 mm.
6. MR-J5W_-G and MR-J5D2-G4 will be supported in the near future.

List of Linear Encoders (Note 1)

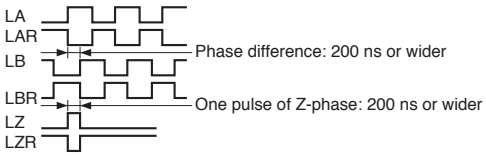
Mitsubishi Electric high-speed serial communication-compatible incremental type

Manufacturer	Model	Resolution	Rated speed (Note 2)	Maximum effective measurement length (Note 3)	Communication method (Note 4)
Magnescale Co., Ltd.	SR75	0.01 µm, 0.05 µm	3.3 m/s	2040 mm	Two-wire type
	SR85			3040 mm	
	SL710 + PL101-RM/RHM	0.1 µm	10.0 m/s	100000 mm	
	SQ10 + PQ10 + MQ10	0.05 µm, 0.1 µm	10.0 m/s	3800 mm	Two-wire type/ Four-wire type
	BL50H + BD700	400 nm/800-fold subdivision (0.5 nm) (Note 11, 12) 400 nm/400-fold subdivision (1 nm) (Note 11)	3.0 m/s	1070 mm	Two-wire type
	BL50H-FSE (Note 10)			1400 mm	
Heidenhain	LIDA 483 + EIB 3091M (16384-fold subdivision) (Note 7)	20 µm/16384-fold subdivision (Approx. 1.22 nm)	4.0 m/s	3040 mm	Four-wire type
	LIDA 485 + EIB 3091M (16384-fold subdivision) (Note 7)			30040 mm	
	LIDA 487 + EIB 3091M (16384-fold subdivision) (Note 7)			6040 mm	
	LIDA 489 + EIB 3091M (16384-fold subdivision) (Note 7)			1020 mm	
	LIDA 287 + EIB 3091M (16384-fold subdivision) (Note 7)	200 µm/16384-fold subdivision (Approx. 12.2 nm)		10000 mm	
	LIDA 289 + EIB 3091M (16384-fold subdivision) (Note 7)				
	LIF 481 + EIB 3091M (4096-fold subdivision)	4 µm/4096-fold subdivision (Approx. 0.977 nm)	1.6 m/s	1640 mm (Note 9)	
	LIP 6081 + EIB 3091M (4096-fold subdivision)			3040 mm (Note 9)	
Nidec Instruments Corporation	PSLH041	0.1 µm	5.0 m/s	2400 mm	Two-wire type
Nidec Machine Tool Corporation	MPFA-HI-M01 (Note 6)	0.1 µm	30.0 m/s	10000 mm (Note 8)	Two-wire type

A/B/Z-phase differential output type (Note 13)

Manufacturer	Model	Resolution	Rated speed (Note 2)	Maximum effective measurement length (Note 3)	Communication method (Note 4)
Not designated	-	0.001 µm to 5 µm (Note 5)	Depends on the linear encoder	Depends on the linear encoder	A/B/Z-phase differential output method

Notes:
1. Contact the relevant linear encoder manufacturer for details on operating environment and specifications of the linear encoder such as ambient temperature, vibration resistance and IP rating.
2. The listed values are the manufacturer's specifications. When combined with servo amplifiers, the specification value is either the listed value or the servo motor maximum speed, whichever is lower.
3. The listed values are the manufacturer's specifications for linear encoder effective measurement lengths. The maximum length of the encoder cable between a linear encoder and a servo amplifier is 30 m. For a linear encoder manufactured by Nidec Machine Tool Corporation, the maximum length of the encoder cable between the linear encoder and a servo amplifier is 20 m.
4. The compatible communication method varies by the servo amplifier and operation mode. Refer to "External Encoder Connection Specifications" in this catalog.
5. Select the linear encoder within this range.
6. There are some restrictions on this linear encoder. When using it, contact your local sales office.
7. For this combination, it is recommended using EIB 3091M with a subdivision of 16384. EIB 3091M with a subdivision of 4096 is also available. Contact the manufacturer for details.
8. For the measurement length over 10000 mm, contact Nidec Machine Tool Corporation.
9. The maximum effective measurement length varies depending on the scale to be used. Contact the manufacturer for details.
10. This linear encoder does not support a fully closed loop control system. In addition, it cannot be used with 3-axis servo amplifiers and 3-axis drive units.
11. The supported resolution is 1 nm or 0.5 nm. The resolution is 1 nm for 400-fold subdivision, and it is 0.5 nm for 800-fold subdivision. Contact the manufacturer for details.
12. The LM-H4M series, LM-U2 (medium thrust) series, and LM-AJ series linear servo motors support a resolution of 0.5 nm.
13. The phase difference of the A-phase pulse and the B-phase pulse, and the width of the Z-phase pulse must be 200 ns or wider. The output pulse of A-phase and B-phase of the A/B/Z-phase differential output linear encoder is in the multiply-by-four count method. For linear encoders without Z-phase, some of the homing modes cannot be used. Refer to "MR-J5 User's Manual" for details.



Drive units

Item		Model	Rated output	Main circuit power supply
Drive unit MR-J5D1-G4	400 V	MR-J5D1-100G4	1 kW	Main circuit power is supplied from the power regeneration converter unit to the drive unit.
		MR-J5D1-200G4	2 kW	
		MR-J5D1-350G4	3.5 kW	
		MR-J5D1-500G4	5 kW	
		MR-J5D1-700G4	7 kW	
Drive unit MR-J5D2-G4	400 V	MR-J5D2-100G4	1 kW x 2 axes	Main circuit power is supplied from the power regeneration converter unit to the drive unit.
		MR-J5D2-200G4	2 kW x 2 axes	
		MR-J5D2-350G4	3.5 kW x 2 axes	
		MR-J5D2-500G4	5 kW x 2 axes	
		MR-J5D2-700G4	7 kW x 2 axes	
Drive unit MR-J5D3-G4	400 V	MR-J5D3-100G4	1 kW x 3 axes	Main circuit power is supplied from the power regeneration converter unit to the drive unit.
		MR-J5D3-200G4	2 kW x 3 axes	
Drive unit MR-J5D1-G4-N1	400 V	MR-J5D1-100G4-N1	1 kW	Main circuit power is supplied from the power regeneration converter unit to the drive unit.
		MR-J5D1-200G4-N1	2 kW	
		MR-J5D1-350G4-N1	3.5 kW	
		MR-J5D1-500G4-N1	5 kW	
		MR-J5D1-700G4-N1	7 kW	
Drive unit MR-J5D2-G4-N1	400 V	MR-J5D2-100G4-N1	1 kW x 2 axes	Main circuit power is supplied from the power regeneration converter unit to the drive unit.
		MR-J5D2-200G4-N1	2 kW x 2 axes	
		MR-J5D2-350G4-N1	3.5 kW x 2 axes	
		MR-J5D2-500G4-N1	5 kW x 2 axes	
		MR-J5D2-700G4-N1	7 kW x 2 axes	
Drive unit MR-J5D3-G4-N1	400 V	MR-J5D3-100G4-N1	1 kW x 3 axes	Main circuit power is supplied from the power regeneration converter unit to the drive unit.
		MR-J5D3-200G4-N1	2 kW x 3 axes	

Board-type servo amplifier

Item		Model	Rated output	Main circuit power supply
Board-type servo amplifier MR-MD	48 V	MR-MD333G	30 W	48 V DC
		MR-MD333G-N1	30 W	48 V DC

Rotary servo motors

Item		Flange size [mm]	Model	Rated output	Rated speed
HK-KT series B: With an electromagnetic brake	HK-KT_W	40 x 40	HK-KT053W(B)	0.05 kW	3000 r/min
			HK-KT13W(B)	0.1 kW	3000 r/min
			HK-KT1M3W(B)	0.15 kW	3000 r/min
		60 x 60	HK-KT13UW(B)	0.1 kW	3000 r/min
			HK-KT23W(B)	0.2 kW	3000 r/min
			HK-KT43W(B)	0.4 kW	3000 r/min
			HK-KT63W(B)	0.6 kW	3000 r/min
		80 x 80	HK-KT23UW(B)	0.2 kW	3000 r/min
			HK-KT43UW(B)	0.4 kW	3000 r/min
			HK-KT7M3W(B)	0.75 kW	3000 r/min
			HK-KT103W(B)	1.0 kW	3000 r/min
		90 x 90	HK-KT63UW(B)	0.6 kW	3000 r/min
			HK-KT7M3UW(B)	0.75 kW	3000 r/min
			HK-KT103UW(B)	1.0 kW	3000 r/min
			HK-KT153W(B)	1.5 kW	3000 r/min
			HK-KT203W(B)	2.0 kW	3000 r/min
			HK-KT202W(B)	2.0 kW	2000 r/min
	HK-KT_4_W	60 x 60	HK-KT434W(B)	0.4 kW	3000 r/min
			HK-KT634W(B)	0.6 kW	3000 r/min
		80 x 80	HK-KT7M34W(B)	0.75 kW	3000 r/min
			HK-KT1034W(B)	1.0 kW	3000 r/min
		90 x 90	HK-KT634UW(B)	0.6 kW	3000 r/min
			HK-KT1034UW(B)	1.0 kW	3000 r/min
			HK-KT1534W(B)	1.5 kW	3000 r/min
			HK-KT2034W(B)	2.0 kW	3000 r/min
			HK-KT2024W(B)	2.0 kW	2000 r/min
Servo motors with functional safety HK-KT series B: With an electromagnetic brake	HK-KT_W_WS	40 x 40	HK-KT053W(B)WS	0.05 kW	3000 r/min
			HK-KT13W(B)WS	0.1 kW	3000 r/min
			HK-KT1M3W(B)WS	0.15 kW	3000 r/min
		60 x 60	HK-KT13UW(B)WS	0.1 kW	3000 r/min
			HK-KT23W(B)WS	0.2 kW	3000 r/min
			HK-KT43W(B)WS	0.4 kW	3000 r/min
			HK-KT63W(B)WS	0.6 kW	3000 r/min
		80 x 80	HK-KT23UW(B)WS	0.2 kW	3000 r/min
			HK-KT43UW(B)WS	0.4 kW	3000 r/min
			HK-KT7M3W(B)WS	0.75 kW	3000 r/min
			HK-KT103W(B)WS	1.0 kW	3000 r/min
		90 x 90	HK-KT63UW(B)WS	0.6 kW	3000 r/min
			HK-KT7M3UW(B)WS	0.75 kW	3000 r/min
			HK-KT103UW(B)WS	1.0 kW	3000 r/min
			HK-KT153W(B)WS	1.5 kW	3000 r/min
			HK-KT203W(B)WS	2.0 kW	3000 r/min
			HK-KT202W(B)WS	2.0 kW	2000 r/min
	HK-KT_4_W_WS	60 x 60	HK-KT434W(B)WS	0.4 kW	3000 r/min
			HK-KT634W(B)WS	0.6 kW	3000 r/min
		80 x 80	HK-KT7M34W(B)WS	0.75 kW	3000 r/min
			HK-KT1034W(B)WS	1.0 kW	3000 r/min
		90 x 90	HK-KT634UW(B)WS	0.6 kW	3000 r/min
			HK-KT1034UW(B)WS	1.0 kW	3000 r/min
			HK-KT1534W(B)WS	1.5 kW	3000 r/min
			HK-KT2034W(B)WS	2.0 kW	3000 r/min
			HK-KT2024W(B)WS	2.0 kW	2000 r/min
Board-type servo amplifier complatible servo motors HK-KT series	HK-KT_	40 x 40	HK-KT0536E2-S1	0.05 kW	3000 r/min

Rotary servo motors

Item		Model		Rated output	Rated speed	Reduction ratio
HK-KT series With a shaft-output type gear reducer for high precision applications, flange mounting B: With an electromagnetic brake	HK-KT_G_	HK-KT053(B)G7	1/5 (40 x 40)	0.05 kW	3000 r/min	1/5 (flange dimensions: 40 mm x 40 mm)
		HK-KT053(B)G7	1/5 (60 x 60)	0.05 kW	3000 r/min	1/5 (flange dimensions: 60 mm x 60 mm)
		HK-KT053(B)G7	1/9	0.05 kW	3000 r/min	1/9
		HK-KT053(B)G7	1/11	0.05 kW	3000 r/min	1/11
		HK-KT053(B)G7	1/21	0.05 kW	3000 r/min	1/21
		HK-KT053(B)G7	1/33	0.05 kW	3000 r/min	1/33
		HK-KT053(B)G7	1/45	0.05 kW	3000 r/min	1/45
		HK-KT13(B)G7	1/5 (40 x 40)	0.1 kW	3000 r/min	1/5 (flange dimensions: 40 mm x 40 mm)
		HK-KT13(B)G7	1/5 (60 x 60)	0.1 kW	3000 r/min	1/5 (flange dimensions: 60 mm x 60 mm)
		HK-KT13(B)G7	1/11	0.1 kW	3000 r/min	1/11
		HK-KT13(B)G7	1/21	0.1 kW	3000 r/min	1/21
		HK-KT13(B)G7	1/33	0.1 kW	3000 r/min	1/33
		HK-KT13(B)G7	1/45	0.1 kW	3000 r/min	1/45
		HK-KT23(B)G7	1/5	0.2 kW	3000 r/min	1/5
		HK-KT23(B)G7	1/11	0.2 kW	3000 r/min	1/11
		HK-KT23(B)G7	1/21	0.2 kW	3000 r/min	1/21
		HK-KT23(B)G7	1/33	0.2 kW	3000 r/min	1/33
		HK-KT23(B)G7	1/45	0.2 kW	3000 r/min	1/45
		HK-KT43(B)G7	1/5	0.4 kW	3000 r/min	1/5
		HK-KT43(B)G7	1/11	0.4 kW	3000 r/min	1/11
		HK-KT43(B)G7	1/21	0.4 kW	3000 r/min	1/21
		HK-KT43(B)G7	1/33	0.4 kW	3000 r/min	1/33
		HK-KT43(B)G7	1/45	0.4 kW	3000 r/min	1/45
		HK-KT7M3(B)G7	1/5	0.75 kW	3000 r/min	1/5
		HK-KT7M3(B)G7	1/11	0.75 kW	3000 r/min	1/11
		HK-KT7M3(B)G7	1/21	0.75 kW	3000 r/min	1/21
		HK-KT7M3(B)G7	1/33	0.75 kW	3000 r/min	1/33
		HK-KT7M3(B)G7	1/45	0.75 kW	3000 r/min	1/45

Rotary servo motors

Item		Flange size [mm]	Model	Rated output	Rated speed
HK-ST series B: With an electromagnetic brake	HK-ST_W	130 x 130	HK-ST52W(B)	0.5 kW	2000 r/min
			HK-ST102W(B)	1.0 kW	2000 r/min
			HK-ST172W(B)	1.75 kW	2000 r/min
			HK-ST202AW(B)	2.0 kW	2000 r/min
			HK-ST302W(B)	3.0 kW	2000 r/min
			HK-ST353W(B)	3.5 kW	3000 r/min
			HK-ST503W(B)	5.0 kW	3000 r/min
		176 x 176	HK-ST7M2UW(B)	0.75 kW	2000 r/min
			HK-ST172UW(B)	1.75 kW	2000 r/min
			HK-ST202W(B)	2.0 kW	2000 r/min
			HK-ST352W(B)	3.5 kW	2000 r/min
			HK-ST502W(B)	5.0 kW	2000 r/min
			HK-ST702W(B)	7.0 kW	2000 r/min
			HK-ST703W(B)	7.0 kW	3000 r/min
			HK-ST903W(B)	9.0 kW	3000 r/min
	HK-ST_4_W	130 x 130	HK-ST524W(B)	0.5 kW	2000 r/min
			HK-ST1024W(B)	1.0 kW	2000 r/min
			HK-ST1724W(B)	1.75 kW	2000 r/min
			HK-ST2024AW(B)	2.0 kW	2000 r/min
			HK-ST3024W(B)	3.0 kW	2000 r/min
			HK-ST3534W(B)	3.5 kW	3000 r/min
			HK-ST5034W(B)	5.0 kW	3000 r/min
			HK-ST7034W(B)	7.0 kW	3000 r/min
			HK-ST9034W(B)	9.0 kW	3000 r/min
		176 x 176	HK-ST2024W(B)	2.0 kW	2000 r/min
			HK-ST3524W(B)	3.5 kW	2000 r/min
			HK-ST5024W(B)	5.0 kW	2000 r/min
			HK-ST7024W(B)	7.0 kW	2000 r/min
Servo motors with functional safety HK-ST series B: With an electromagnetic brake	HK-ST_W_WS	130 x 130	HK-ST52W(B)WS	0.5 kW	2000 r/min
			HK-ST102W(B)WS	1.0 kW	2000 r/min
			HK-ST172W(B)WS	1.75 kW	2000 r/min
			HK-ST202AW(B)WS	2.0 kW	2000 r/min
			HK-ST302W(B)WS	3.0 kW	2000 r/min
			HK-ST353W(B)WS	3.5 kW	3000 r/min
			HK-ST503W(B)WS	5.0 kW	3000 r/min
		176 x 176	HK-ST7M2UW(B)WS	0.75 kW	2000 r/min
			HK-ST172UW(B)WS	1.75 kW	2000 r/min
			HK-ST202W(B)WS	2.0 kW	2000 r/min
			HK-ST352W(B)WS	3.5 kW	2000 r/min
			HK-ST502W(B)WS	5.0 kW	2000 r/min
			HK-ST702W(B)WS	7.0 kW	2000 r/min
	HK-ST_4_W_WS	130 x 130	HK-ST524W(B)WS	0.5 kW	2000 r/min
			HK-ST1024W(B)WS	1.0 kW	2000 r/min
			HK-ST1724W(B)WS	1.75 kW	2000 r/min
			HK-ST2024AW(B)WS	2.0 kW	2000 r/min
			HK-ST3024W(B)WS	3.0 kW	2000 r/min
			HK-ST3534W(B)WS	3.5 kW	3000 r/min
			HK-ST5034W(B)WS	5.0 kW	3000 r/min
		176 x 176	HK-ST2024W(B)WS	2.0 kW	2000 r/min
			HK-ST3524W(B)WS	3.5 kW	2000 r/min
			HK-ST5024W(B)WS	5.0 kW	2000 r/min
			HK-ST7024W(B)WS	7.0 kW	2000 r/min

Rotary servo motors

Item		Model	Rated output	Rated speed	Reduction ratio
HK-ST series With a gear reducer for general industrial machines B: With an electromagnetic brake G1: Flange mounting G1H: Foot mounting	HK-ST_4_G_	HK-ST524(B)G1(H) 1/6	0.5 kW	2000 r/min	1/6
		HK-ST524(B)G1(H) 1/11	0.5 kW	2000 r/min	1/11
		HK-ST524(B)G1(H) 1/17	0.5 kW	2000 r/min	1/17
		HK-ST524(B)G1(H) 1/29	0.5 kW	2000 r/min	1/29
		HK-ST524(B)G1(H) 1/35	0.5 kW	2000 r/min	1/35
		HK-ST524(B)G1(H) 1/43	0.5 kW	2000 r/min	1/43
		HK-ST524(B)G1(H) 1/59	0.5 kW	2000 r/min	1/59
		HK-ST1024(B)G1(H) 1/6	1.0 kW	2000 r/min	1/6
		HK-ST1024(B)G1(H) 1/11	1.0 kW	2000 r/min	1/11
		HK-ST1024(B)G1(H) 1/17	1.0 kW	2000 r/min	1/17
		HK-ST1024(B)G1(H) 1/29	1.0 kW	2000 r/min	1/29
		HK-ST1024(B)G1(H) 1/35	1.0 kW	2000 r/min	1/35
		HK-ST1024(B)G1(H) 1/43	1.0 kW	2000 r/min	1/43
		HK-ST1024(B)G1(H) 1/59	1.0 kW	2000 r/min	1/59
		HK-ST1524(B)G1(H) 1/6	1.5 kW	2000 r/min	1/6
		HK-ST1524(B)G1(H) 1/11	1.5 kW	2000 r/min	1/11
		HK-ST1524(B)G1(H) 1/17	1.5 kW	2000 r/min	1/17
		HK-ST1524(B)G1(H) 1/29	1.5 kW	2000 r/min	1/29
		HK-ST1524(B)G1(H) 1/35	1.5 kW	2000 r/min	1/35
		HK-ST1524(B)G1(H) 1/43	1.5 kW	2000 r/min	1/43
		HK-ST1524(B)G1(H) 1/59	1.5 kW	2000 r/min	1/59
		HK-ST2024(B)G1(H) 1/6	2.0 kW	2000 r/min	1/6
		HK-ST2024(B)G1(H) 1/11	2.0 kW	2000 r/min	1/11
		HK-ST2024(B)G1(H) 1/17	2.0 kW	2000 r/min	1/17
		HK-ST2024(B)G1(H) 1/29	2.0 kW	2000 r/min	1/29
		HK-ST2024(B)G1(H) 1/35	2.0 kW	2000 r/min	1/35
		HK-ST2024(B)G1(H) 1/43	2.0 kW	2000 r/min	1/43
		HK-ST2024(B)G1(H) 1/59	2.0 kW	2000 r/min	1/59
		HK-ST3524(B)G1(H) 1/6	3.5 kW	2000 r/min	1/6
		HK-ST3524(B)G1(H) 1/11	3.5 kW	2000 r/min	1/11
		HK-ST3524(B)G1(H) 1/17	3.5 kW	2000 r/min	1/17
		HK-ST3524(B)G1(H) 1/29	3.5 kW	2000 r/min	1/29
		HK-ST3524(B)G1(H) 1/35	3.5 kW	2000 r/min	1/35
		HK-ST3524(B)G1(H) 1/43	3.5 kW	2000 r/min	1/43
		HK-ST3524(B)G1(H) 1/59	3.5 kW	2000 r/min	1/59
		HK-ST5024(B)G1(H) 1/6	5.0 kW	2000 r/min	1/6
		HK-ST5024(B)G1(H) 1/11	5.0 kW	2000 r/min	1/11
		HK-ST5024(B)G1(H) 1/17	5.0 kW	2000 r/min	1/17
		HK-ST5024(B)G1(H) 1/29	5.0 kW	2000 r/min	1/29
		HK-ST5024(B)G1(H) 1/35	5.0 kW	2000 r/min	1/35
		HK-ST5024(B)G1(H) 1/43	5.0 kW	2000 r/min	1/43
		HK-ST5024(B)G1(H) 1/59	5.0 kW	2000 r/min	1/59
		HK-ST7024(B)G1(H) 1/6	7.0 kW	2000 r/min	1/6
		HK-ST7024(B)G1(H) 1/11	7.0 kW	2000 r/min	1/11
		HK-ST7024(B)G1(H) 1/17	7.0 kW	2000 r/min	1/17
		HK-ST7024(B)G1(H) 1/29	7.0 kW	2000 r/min	1/29
		HK-ST7024(B)G1(H) 1/35	7.0 kW	2000 r/min	1/35
		HK-ST7024(B)G1(H) 1/43	7.0 kW	2000 r/min	1/43
		HK-ST7024(B)G1(H) 1/59	7.0 kW	2000 r/min	1/59

Rotary servo motors

Item		Model		Rated output	Rated speed	Reduction ratio
HK-ST series With a shaft-output type gear reducer for high precision applications, flange mounting B: With an electromagnetic brake	HK-ST_G_	HK-ST52(B)G7	1/5	0.5 kW	2000 r/min	1/5
		HK-ST52(B)G7	1/11	0.5 kW	2000 r/min	1/11
		HK-ST52(B)G7	1/21	0.5 kW	2000 r/min	1/21
		HK-ST52(B)G7	1/33	0.5 kW	2000 r/min	1/33
		HK-ST52(B)G7	1/45	0.5 kW	2000 r/min	1/45
		HK-ST102(B)G7	1/5	1.0 kW	2000 r/min	1/5
		HK-ST102(B)G7	1/11	1.0 kW	2000 r/min	1/11
		HK-ST102(B)G7	1/21	1.0 kW	2000 r/min	1/21
		HK-ST102(B)G7	1/33	1.0 kW	2000 r/min	1/33
		HK-ST102(B)G7	1/45	1.0 kW	2000 r/min	1/45
		HK-ST152(B)G7	1/5	1.5 kW	2000 r/min	1/5
		HK-ST152(B)G7	1/11	1.5 kW	2000 r/min	1/11
		HK-ST152(B)G7	1/21	1.5 kW	2000 r/min	1/21
		HK-ST152(B)G7	1/33	1.5 kW	2000 r/min	1/33
		HK-ST152(B)G7	1/45	1.5 kW	2000 r/min	1/45
		HK-ST202(B)G7	1/5	2.0 kW	2000 r/min	1/5
		HK-ST202(B)G7	1/11	2.0 kW	2000 r/min	1/11
		HK-ST202(B)G7	1/21	2.0 kW	2000 r/min	1/21
		HK-ST202(B)G7	1/33	2.0 kW	2000 r/min	1/33
		HK-ST202(B)G7	1/45	2.0 kW	2000 r/min	1/45
		HK-ST352(B)G7	1/5	3.5 kW	2000 r/min	1/5
		HK-ST352(B)G7	1/11	3.5 kW	2000 r/min	1/11
		HK-ST352(B)G7	1/21	3.5 kW	2000 r/min	1/21
		HK-ST502(B)G7	1/5	5.0 kW	2000 r/min	1/5
		HK-ST502(B)G7	1/11	5.0 kW	2000 r/min	1/11
		HK-ST702(B)G7	1/5	7.0 kW	2000 r/min	1/5
	HK-ST_4_G_	HK-ST524(B)G7	1/5	0.5 kW	2000 r/min	1/5
		HK-ST524(B)G7	1/11	0.5 kW	2000 r/min	1/11
		HK-ST524(B)G7	1/21	0.5 kW	2000 r/min	1/21
		HK-ST524(B)G7	1/33	0.5 kW	2000 r/min	1/33
		HK-ST524(B)G7	1/45	0.5 kW	2000 r/min	1/45
		HK-ST1024(B)G7	1/5	1.0 kW	2000 r/min	1/5
		HK-ST1024(B)G7	1/11	1.0 kW	2000 r/min	1/11
		HK-ST1024(B)G7	1/21	1.0 kW	2000 r/min	1/21
		HK-ST1024(B)G7	1/33	1.0 kW	2000 r/min	1/33
		HK-ST1024(B)G7	1/45	1.0 kW	2000 r/min	1/45
		HK-ST1524(B)G7	1/5	1.5 kW	2000 r/min	1/5
		HK-ST1524(B)G7	1/11	1.5 kW	2000 r/min	1/11
		HK-ST1524(B)G7	1/21	1.5 kW	2000 r/min	1/21
		HK-ST1524(B)G7	1/33	1.5 kW	2000 r/min	1/33
		HK-ST1524(B)G7	1/45	1.5 kW	2000 r/min	1/45
		HK-ST2024(B)G7	1/5	2.0 kW	2000 r/min	1/5
		HK-ST2024(B)G7	1/11	2.0 kW	2000 r/min	1/11
		HK-ST2024(B)G7	1/21	2.0 kW	2000 r/min	1/21
		HK-ST2024(B)G7	1/33	2.0 kW	2000 r/min	1/33
		HK-ST2024(B)G7	1/45	2.0 kW	2000 r/min	1/45
		HK-ST3524(B)G7	1/5	3.5 kW	2000 r/min	1/5
		HK-ST3524(B)G7	1/11	3.5 kW	2000 r/min	1/11
		HK-ST3524(B)G7	1/21	3.5 kW	2000 r/min	1/21
		HK-ST5024(B)G7	1/5	5.0 kW	2000 r/min	1/5
		HK-ST5024(B)G7	1/11	5.0 kW	2000 r/min	1/11
		HK-ST7024(B)G7	1/5	7.0 kW	2000 r/min	1/5

Direct drive motors

Item	Model	Rated torque	Maximum torque	Rated speed
TM-RG2M series	TM-RG2M002C30	2.2 N•m	8.8 N•m	300 r/min
	TM-RG2M004E30	4.5 N•m	13.5 N•m	300 r/min
	TM-RG2M009G30	9 N•m	27 N•m	300 r/min
TM-RU2M series	TM-RU2M002C30	2.2 N•m	8.8 N•m	300 r/min
	TM-RU2M004E30	4.5 N•m	13.5 N•m	300 r/min
	TM-RU2M009G30	9 N•m	27 N•m	300 r/min
TM-RFM series	TM-RFM002C20	2 N•m	6 N•m	200 r/min
	TM-RFM004C20	4 N•m	12 N•m	200 r/min
	TM-RFM006C20	6 N•m	18 N•m	200 r/min
	TM-RFM006E20	6 N•m	18 N•m	200 r/min
	TM-RFM012E20	12 N•m	36 N•m	200 r/min
	TM-RFM018E20	18 N•m	54 N•m	200 r/min
	TM-RFM012G20	12 N•m	36 N•m	200 r/min
	TM-RFM048G20	48 N•m	144 N•m	200 r/min
	TM-RFM072G20	72 N•m	216 N•m	200 r/min
	TM-RFM040J10	40 N•m	120 N•m	100 r/min
	TM-RFM120J10	120 N•m	360 N•m	100 r/min
	TM-RFM240J10	240 N•m	720 N•m	100 r/min